

Identify the Gap

Hydroficient IoT Cyber Defense Externship – Project 5, Step 1

Estimated time to complete: 15 minutes

Instructions: Complete all three parts below. Fill in the blanks, check the boxes that apply, and write your analysis in the spaces provided. Submit this document to the shared Google Doc when finished.

Part 1: Review Your Project 4 Configuration

Open your `mosquitto_tls.conf` from Project 4 and answer these questions.

1. What port is the TLS listener using?

Port: _____

2. What files are specified for the broker's identity?

CA file: _____

Certificate file: _____

Key file: _____

3. Is there a `require_certificate` line?

- ☐ Yes, it's set to true
- ☐ Yes, it's set to false
- ☐ No, this line doesn't exist

4. Based on your configuration, what type of TLS is this?

- ☐ One-Way TLS (server proves identity only)
- ☐ Mutual TLS (both sides prove identity)

Part 2: The Rogue Device Scenario

Scenario: An attacker gains physical access to The Grand Marina's network closet. They connect a Raspberry Pi to the network and try to connect to your MQTT broker.

Answer these questions about your current (Project 4) configuration:

1. Can the attacker's device connect to the broker?

- ☐ Yes
- ☐ No

Explain why:

2. What would the attacker need to successfully connect?

- ☐ Just network access
- ☐ Network access + the CA certificate
- ☐ Network access + CA certificate + device certificate + private key

3. Once connected, what could the attacker do? (Check all that apply)

- ☐ Subscribe to all topics and read sensor data
- ☐ Publish fake sensor readings
- ☐ Send commands to HYDROLOGIC devices
- ☐ Pretend to be any device ID they want
- ☐ Take over legitimate device connections

Part 3: Three Scenarios Where One-Way TLS Fails

List three realistic scenarios where your current one-way TLS configuration would fail to prevent an attack.

Hint: Consider insider threats, compromised devices, rogue hardware, man-in-the-middle attacks, or credential theft.

Scenario 1

Scenario name: _____

Describe the attack:

Why one-way TLS doesn't stop it:

Scenario 2

Scenario name: _____

Describe the attack:

Why one-way TLS doesn't stop it:

Scenario 3

Scenario name: _____

Describe the attack:

Why one-way TLS doesn't stop it:
